

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2022 P2HR Worked Solutions

Jun 2022 | P2HR

31 questions ■ question image followed by worked solution

PREPARED BY

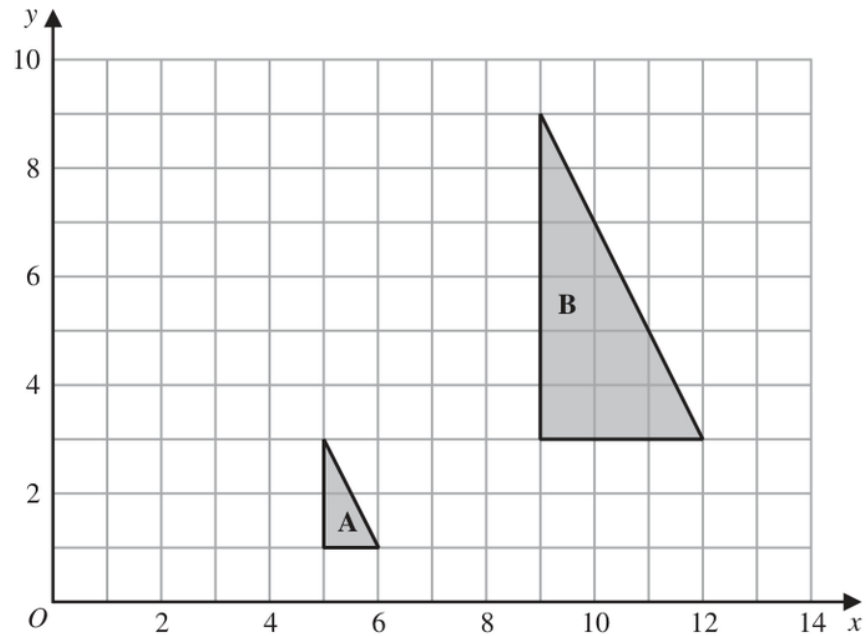
Dr. Eslam Ahmed

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PAST PAPER QUESTION**Q#: 1****Marks: 4****TOPIC: Transformations**

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1

(a) Describe fully the single transformation that maps triangle **A** onto triangle **B**

(3)

(b) On the grid above, translate triangle **A** by the vector $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$

Label your triangle **C**

(1)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1

Marks: 4

TOPIC: **Transformations**

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WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 3****TOPIC: Prime Factors, HCF & LCM**

Jun 2022 P2HR - Jun 2022 | P2HR

- 2** Write 1200 as a product of powers of its prime factors.
Show your working clearly.

(Total for Question 2 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 2

Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 3****Marks: 3****TOPIC: Statistics Toolkit**

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3 Alberto, Bill, Candela and Diana are four friends.

Here is some information about the height of each of these friends.

Alberto's height is 158 cm.

Bill's height is 175 cm.

Candela's height is greater than Diana's height.

The median height of these four friends is 160 cm.

The range of the heights of these four friends is 21 cm.

Work out Candela's height and Diana's height.

Candela cm

Diana cm

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3

Marks: 3

TOPIC: **Statistics Toolkit**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 4****Marks: 5****TOPIC: Set Notation & Venn Diagrams**

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4 $\mathcal{E} = \{9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$ $A = \{\text{multiples of 3}\}$ $B = \{\text{odd numbers}\}$

(a) List the members of the set

(i) $A \cap B$
(1)(ii) $A \cup B$
(1)(b) Is it true that $24 \in A$?

Tick one of the boxes below.

Yes

No

☐☐

Give a reason for your answer.

.....
(1)Set C has 4 members such that $C \cap B' = \{10, 18\}$ (c) List the members of one possible set C
(2)**(Total for Question 4 is 5 marks)**

PAST PAPER SOLUTION

Q#: 4

Marks: 5

TOPIC: **Set Notation & Venn Diagrams**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 5****Marks: 4****TOPIC: Area & Perimeter**

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- 5 The diagram shows a shape made from a square $ABCD$ and 4 identical semicircles.

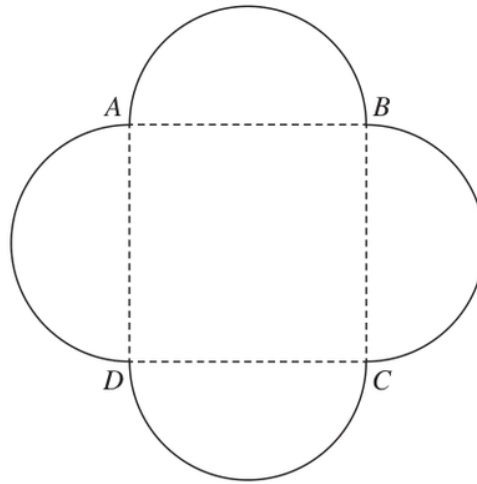


Diagram **NOT**
accurately drawn

As shown in the diagram, the semicircles have AB , BC , CD and DA as diameters.

The area of the square is 36 cm^2

Calculate the total area of the shape.

Give your answer correct to one decimal place.

..... cm^2

(Total for Question 5 is 4 marks)

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Q#: 5

Marks: 4

TOPIC: **Area & Perimeter**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 6****Marks: 8****TOPIC: Solving Linear Equations**

Jun 2022 P2HR - Jun 2022 | P2HR

6 (a) Solve $p = \frac{3p - 5}{10}$

Show clear algebraic working.

$$p = \dots\dots\dots (3)$$

(b) Simplify a^0 where $a > 0$

$$\dots\dots\dots (1)$$

(c) Simplify fully $\frac{3xy^3}{6x^2y}$

$$\dots\dots\dots (2)$$

(d) Factorise fully $10c^3d^2 + 15cd^4$

$$\dots\dots\dots (2)$$

(Total for Question 6 is 8 marks)

PAST PAPER SOLUTION

Q#: 6

Marks: 8

TOPIC: **Solving Linear Equations**

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PAST PAPER QUESTION**Q#: 7****Marks: 2****TOPIC: Algebraic Roots & Indices**

Jun 2022 P2HR - Jun 2022 | P2HR

7 $\frac{2^k}{4^n} = 2^x$

Find an expression for x in terms of k and n

$x = \dots\dots\dots$

(Total for Question 7 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 7

Marks: 2

TOPIC: **Algebraic Roots & Indices**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 8****Marks: 3**TOPIC: **Percentages**

Jun 2022 P2HR - Jun 2022 | P2HR

- 8** A cinema increased the cost of an adult ticket by 12%
- After the increase, the cost of an adult ticket was £18.20
- Work out the cost of an adult ticket before the increase.

£.....

(Total for Question 8 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 8

Marks: 3

TOPIC: Percentages

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PAST PAPER QUESTION**Q#: 9****Marks: 4****TOPIC: Powers, Roots & Standard Form**

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- 9 The table gives information about the population, correct to 2 significant figures, of each of five cities in 2018

City	Population (2018)
Ahmedabad	7.7×10^6
Barcelona	5.5×10^6
Chicago	8.8×10^6
Lagos	1.3×10^7
Tokyo	3.7×10^7

- (a) Write 8.8×10^6 as an ordinary number.

.....
(1)

- (b) Which of these cities had the least population in 2018?

.....
(1)

- (c) Work out the difference between the population of Tokyo and the population of Ahmedabad in 2018

Give your answer in standard form correct to 2 significant figures.

.....
(2)

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION

Q#: 9

Marks: 4

TOPIC: **Powers, Roots & Standard Form**

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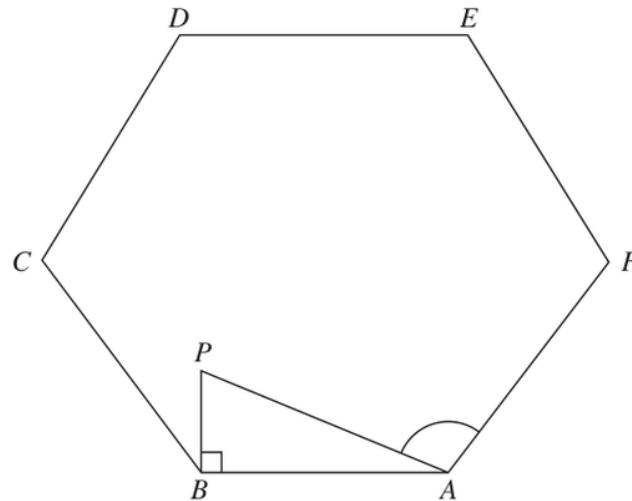
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EA

PAST PAPER QUESTION**Q#: 10****Marks: 5****TOPIC:** Right-Angled Triangles - Pythagoras & Trigonometry

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10 The diagram shows triangle ABP inside the regular hexagon $ABCDEF$ Diagram **NOT**
accurately drawn

$AB = 5 \text{ cm}$

$BP = 2 \text{ cm}$

$\text{Angle } ABP = 90^\circ$

Work out the size of angle PAF

Give your answer correct to 3 significant figures.

o

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10

Marks: 5

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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EA

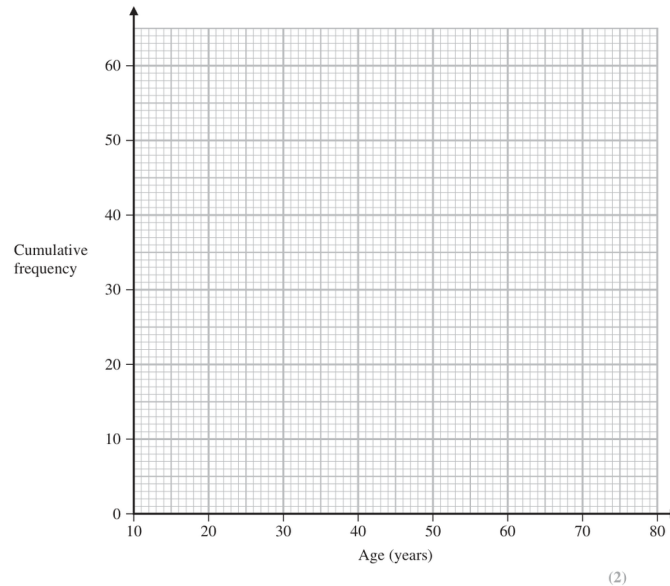
PAST PAPER QUESTION**Q#: 11****Marks: 7****TOPIC: Cumulative Frequency Diagrams**

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- 11 The cumulative frequency table shows information about the ages of 60 people who went to a gym on Saturday.

Age (a years)	Cumulative frequency
$10 < a \leq 20$	13
$10 < a \leq 30$	36
$10 < a \leq 40$	42
$10 < a \leq 50$	47
$10 < a \leq 60$	52
$10 < a \leq 70$	56
$10 < a \leq 80$	60

- (a) On the grid, draw a cumulative frequency graph for the information in the table.

**Question 11 continued**

- (b) Use your graph to find an estimate for the median of the ages of these people.

..... years
(1)

- (c) Use your graph to find an estimate for the interquartile range of the ages of these people.

..... years
(2)

- (d) Use your graph to find an estimate for the number of these people who are older than 55 years.

.....
(2)

(Total for Question 11 is 7 marks)

PAST PAPER SOLUTION

Q#: 11

Marks: 7

TOPIC: **Cumulative Frequency Diagrams**

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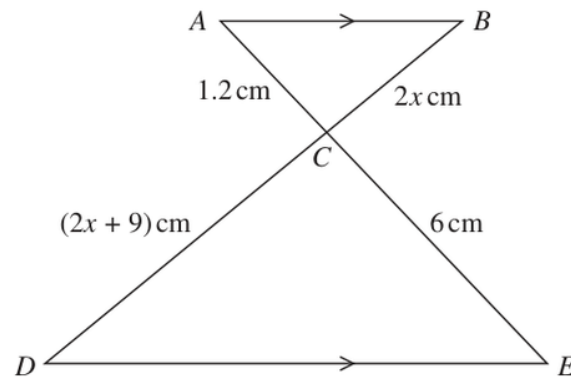
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EA

PAST PAPER QUESTION**Q#: 12****Marks: 3****TOPIC: Congruence, Similarity & Geometrical Proof**

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12Diagram **NOT**
accurately drawn

ACE and BCD are straight lines.
 AB is parallel to DE

Work out the value of x

 $x = \dots\dots\dots$ **(Total for Question 12 is 3 marks)**

PAST PAPER SOLUTION

Q#: 12

Marks: 3

TOPIC: **Congruence, Similarity & Geometrical Proof**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 13****Marks: 3****TOPIC: Circles, Arcs & Sectors**

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13 The diagram shows a sector AOB of a circle with centre O

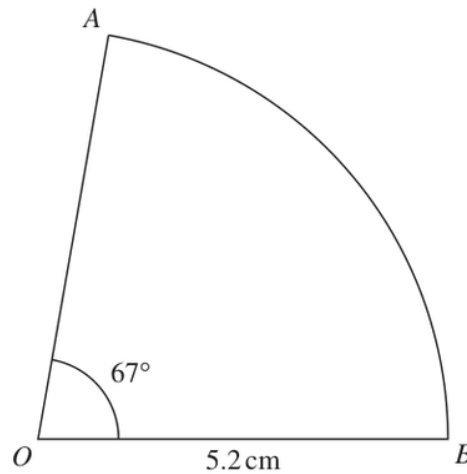


Diagram **NOT**
accurately drawn

Angle $AOB = 67^\circ$
 $OA = OB = 5.2 \text{ cm}$

Calculate the perimeter of the sector.
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13

Marks: 3

TOPIC: **Circles, Arcs & Sectors**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 14****Marks: 3****TOPIC: Combined & Conditional Probability**

Jun 2022 P2HR - Jun 2022 | P2HR

14 Ciara throws **four** fair six-sided dice.

The faces of each dice are labelled with the numbers 1, 2, 3, 4, 5, 6

Work out the probability that at least one of the dice lands on an even number.

(Total for Question 14 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 14

Marks: 3

TOPIC: **Combined & Conditional Probability**

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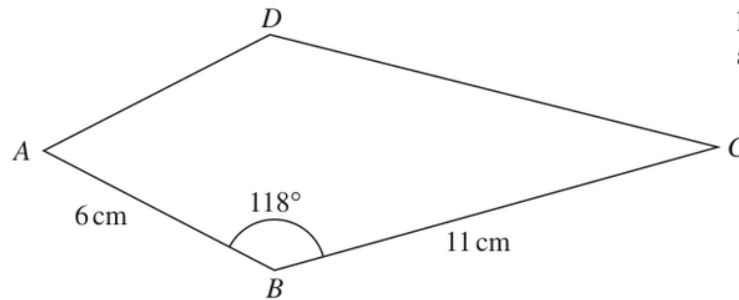
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Sine, Cosine Rule & Area of Triangles**

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15 The diagram shows a kite $ABCD$ 

$AB = 6 \text{ cm}$

$BC = 11 \text{ cm}$

$\text{Angle } ABC = 118^\circ$

Calculate the area of the kite.

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15

Marks: 3

TOPIC: **Sine, Cosine Rule & Area of Triangles**

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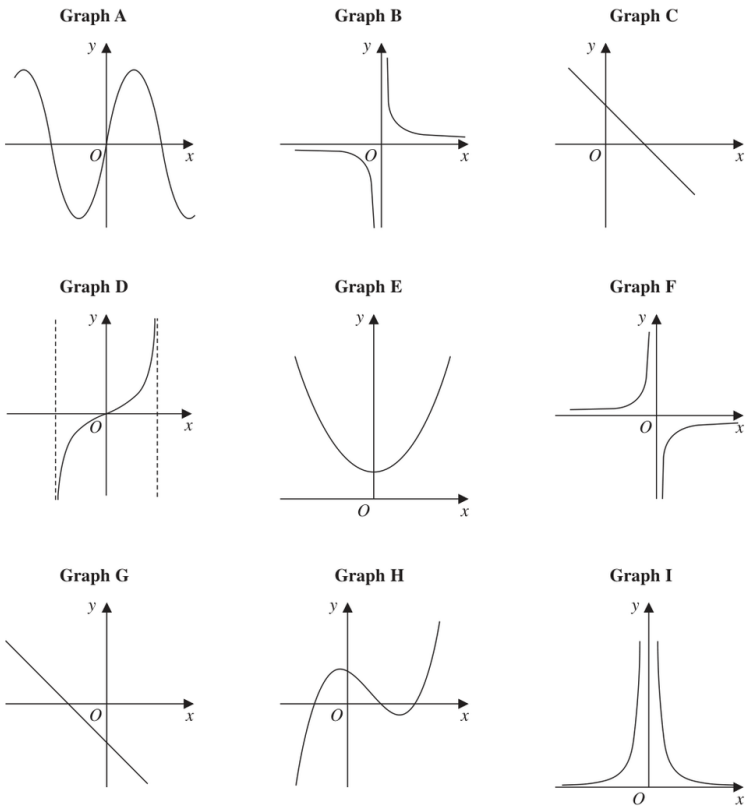
PAST PAPER QUESTION

Q#: 16 Marks: 3

TOPIC: **Graphs of Functions**

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16 Here are nine graphs.



Complete the table below with the letter of the graph that could represent each given equation.
Write each answer on the dotted line.

Equation	Graph
$y = -2x + 3$	
$y = -\frac{1}{x}$	
$y = \tan x^\circ$	
$y = (x + 1)(x - 1)(x - 2)$	

(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION

Q#: 16

Marks: 3

TOPIC: **Graphs of Functions**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 17****Marks: 2**TOPIC: **Algebraic Proof**

Jun 2022 P2HR - Jun 2022 | P2HR

17 Use algebra to show that $0.3\dot{4}\dot{5} = \frac{19}{55}$

(Total for Question 17 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 17

Marks: 2

TOPIC: **Algebraic Proof**

Jun 2022 P2HR - Jun 2022 | P2HR

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 18****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

Jun 2022 P2HR - Jun 2022 | P2HR

18 Kaidan and Sonja went on two different car journeys.

For Kaidan's journey

distance = 80 km correct to the nearest 5 km

time = 2.7 hours correct to 1 decimal place

For Sonja's journey

distance = 33 km correct to 2 significant figures

time = 1 hour correct to the nearest 0.1 hour

Kaidan says,

"My average speed could have been greater than Sonja's average speed."

By considering bounds, show that Kaidan is correct.

Show your working clearly.

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18

Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jun 2022 P2HR - Jun 2022 | P2HR

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 19****Marks: 4****TOPIC: Functions**

Jun 2022 P2HR - Jun 2022 | P2HR

19 $f(x) = x^2 - 4$

$g(x) = 2x + 1$

Solve $fg(x) > 0$

Show clear algebraic working.

(Total for Question 19 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 19

Marks: 4

TOPIC: **Functions**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Coordinate Geometry**

Jun 2022 P2HR - Jun 2022 | P2HR

20 The centre O of a circle has coordinates $(4, 7)$

The point A , on the circle, has coordinates $(6, 11)$ and AOP is a diameter of the circle.

Find an equation of the tangent to the circle at the point P

(Total for Question 20 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 20

Marks: 4

TOPIC: **Coordinate Geometry**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Coordinate Geometry**

Jun 2022 P2HR - Jun 2022 | P2HR

20 The centre O of a circle has coordinates $(4, 7)$

The point A , on the circle, has coordinates $(6, 11)$ and AOP is a diameter of the circle.

Find an equation of the tangent to the circle at the point P

(Total for Question 20 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 20

Marks: 4

TOPIC: **Coordinate Geometry**

Jun 2022 P2HR - Jun 2022 | P2HR

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Simultaneous Equations**

Jun 2022 P2HR - Jun 2022 | P2HR

21 Solve the simultaneous equations

$$\begin{aligned}x - 2y &= 3 \\ x^2 - y^2 + 2x &= 10\end{aligned}$$

Show clear algebraic working.

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 21

Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2022 P2HR - Jun 2022 | P2HR

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Simultaneous Equations**

Jun 2022 P2HR - Jun 2022 | P2HR

21 Solve the simultaneous equations

$$\begin{aligned}x - 2y &= 3 \\ x^2 - y^2 + 2x &= 10\end{aligned}$$

Show clear algebraic working.

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 21

Marks: 5

TOPIC: **Simultaneous Equations**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Transformations of Graphs**

Jun 2022 P2HR - Jun 2022 | P2HR

22 The point A with coordinates $(-3, 2)$ lies on the straight line with equation $y = f(x)$

(a) Find the coordinates of the image of the point A on the straight line with equation

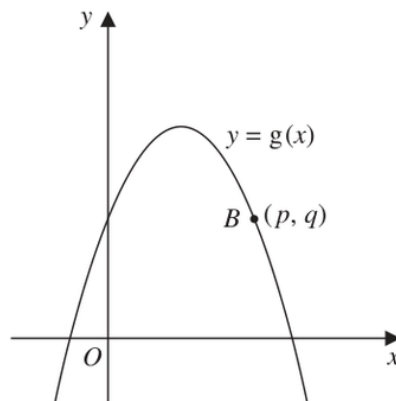
(i) $y = f(x) - 3$

(.....,)
(1)

(ii) $y = f\left(\frac{x}{2}\right)$

(.....,)
(1)

Here is a sketch of part of the curve with equation $y = g(x)$



The point B with coordinates (p, q) lies on the curve.

(b) Find the coordinates of the image of the point B on the curve with equation

$$y = -g(x - c)$$

where c is a constant.

(.....,)
(2)

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22

Marks: 4

TOPIC: **Transformations of Graphs**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Transformations of Graphs**

Jun 2022 P2HR - Jun 2022 | P2HR

22 The point A with coordinates $(-3, 2)$ lies on the straight line with equation $y = f(x)$

(a) Find the coordinates of the image of the point A on the straight line with equation

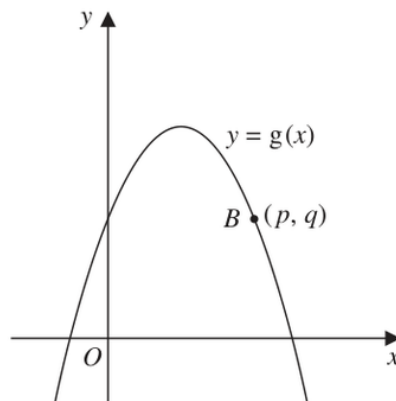
(i) $y = f(x) - 3$

(.....,)
(1)

(ii) $y = f\left(\frac{x}{2}\right)$

(.....,)
(1)

Here is a sketch of part of the curve with equation $y = g(x)$



The point B with coordinates (p, q) lies on the curve.

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$$y = -g(x - c)$$

where c is a constant.

(.....,)
(2)

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22

Marks: 4

TOPIC: **Transformations of Graphs**

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PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Algebraic Fractions**

Jun 2022 P2HR - Jun 2022 | P2HR

23 Express $\left(\frac{20}{x^2 - 36} - \frac{2}{x - 6}\right) \times \frac{1}{4 - x}$ as a single fraction in its simplest form.

(Total for Question 23 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 23

Marks: 3

TOPIC: **Algebraic Fractions**

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EA

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Algebraic Fractions**

Jun 2022 P2HR - Jun 2022 | P2HR

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(Total for Question 23 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 23

Marks: 3

TOPIC: **Algebraic Fractions**

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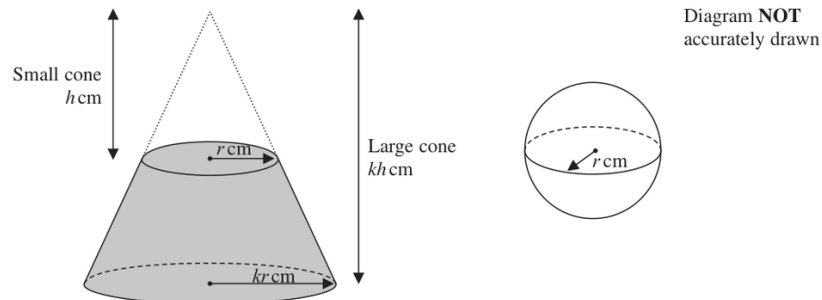
PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Area & Volume of Similar Shapes**

Jun 2022 P2HR - Jun 2022 | P2HR

24 The diagram shows a frustum of a cone, and a sphere.

The frustum, shown shaded in the diagram, is made by removing the small cone from the large cone.

The small cone and the large cone are similar.



The height of the small cone is h cm and the radius of the base of the small cone is r cm.
The height of the large cone is kh cm and the radius of the base of the large cone is kr cm.
The radius of the sphere is r cm.

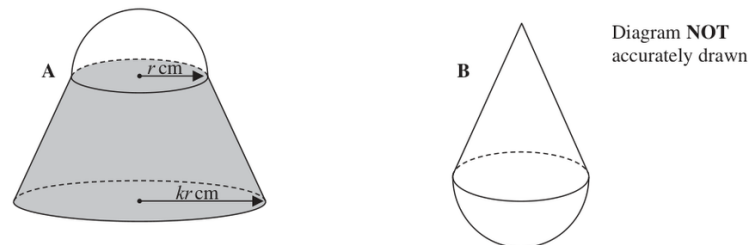
The sphere is divided into two hemispheres, each of radius r cm.

Solid A is formed by joining one of the hemispheres to the frustum.

The plane face of the hemisphere coincides with the upper plane face of the frustum, as shown in the diagram below.

Solid B is formed by joining the other hemisphere to the small cone that was removed from the large cone.

The plane face of the hemisphere coincides with the plane face of the base of the small cone, as shown in the diagram below.



The volume of solid A is 6 times the volume of solid B.

Given that $k > \sqrt[3]{7}$

find an expression for h in terms of k and r

$h = \dots\dots\dots$

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24

Marks: 6

TOPIC: **Area & Volume of Similar Shapes**

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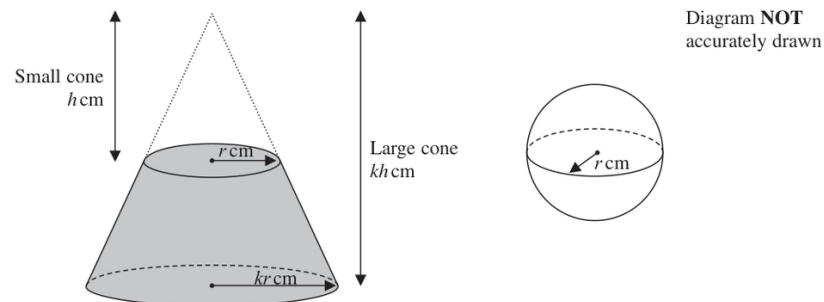
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Jun 2022 P2HR - Jun 2022 | P2HR

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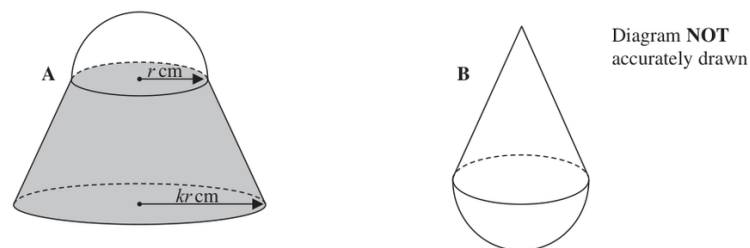
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Given that $k > \sqrt[3]{7}$

find an expression for h in terms of k and r

$h = \dots\dots\dots$

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24

Marks: 6

TOPIC: **Area & Volume of Similar Shapes**

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WORKED SOLUTION

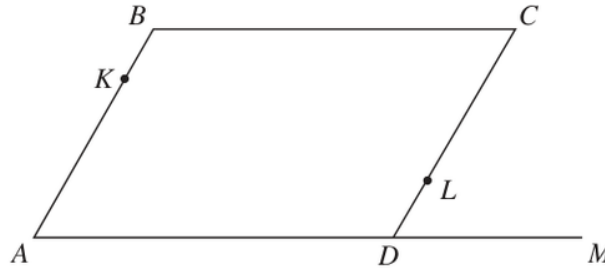
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 25****Marks: 5****TOPIC: Vectors**

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25 $ABCD$ is a parallelogram and ADM is a straight line.Diagram **NOT**
accurately drawn

$$\vec{AB} = \mathbf{a} \quad \vec{BC} = \mathbf{b} \quad \vec{DM} = \frac{1}{2} \mathbf{b}$$

 K is the point on AB such that $AK:AB = \lambda:1$ L is the point on CD such that $CL:CD = \mu:1$ KLM is a straight line.Given that $\lambda:\mu = 1:2$ use a vector method to find the value of λ and the value of μ

$$\lambda = \dots\dots\dots$$

$$\mu = \dots\dots\dots$$

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION

Q#: 25

Marks: 5

TOPIC: **Vectors**

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WORKED SOLUTION

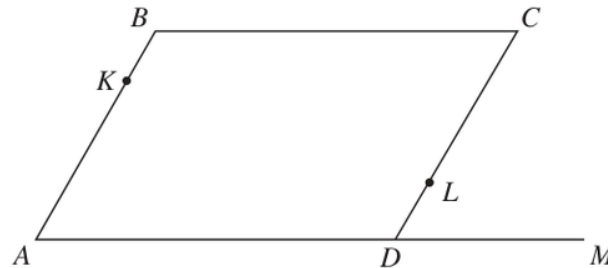
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WORKED SOLUTION

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EA